BetterSplit: A Desktop App for Managing Shared Expenses

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***Abstract*—BetterSplit is a desktop application designed to simplify the management of shared expenses among friends, roommates, and colleagues. This paper presents a revised development report following a full codebase rewrite. The application now includes reimplemented core features from Iteration 1 and an enhanced "Settle Up" feature from Iteration 2. The new architecture improves maintainability, modularity, and user experience.**

***Index Terms*—Expense sharing, desktop application, WPF, real-time syncing, debt tracking**

I. Introduction

Managing shared expenses often becomes a chaotic task when done manually or using basic tools. Miscommunication, unclear records, and manual calculations lead to confusion and disputes. BetterSplit addresses these issues by offering an automated, transparent, and easy-to-use expense management system. After the initial implementation, the development team opted for a complete rewrite to improve scalability, code clarity, and user interaction. This report outlines the restructured development process, focusing on reimplementation of Iteration 1 features and the enhanced integration of the "Settle Up" module from Iteration 2.

II. Rewritten Application Architecture

The app was entirely restructured in terms of backend logic, UI layout, and modular code organization. Built using Windows Presentation Foundation (WPF) in C#, the new version introduces an MVVM-based design pattern for clean separation between the interface and logic.

A. Data Models

New data models were introduced for expenses, users, and groups. These models are now more adaptable, supporting future features such as advanced filtering and user roles.

B. UI Enhancements

The user interface was redesigned for a cleaner layout and smoother interactions. Navigation between dashboards, group views, and individual expenses is more intuitive.

C. Improved State Management

Real-time updates and user state changes are now efficiently managed using centralized services that handle data binding across views.

III. Feature Reimplementation: Iteration 1

The first development sprint focused on restoring and improving the foundational features:

- **User Authentication:** A simplified login system was redeveloped with improved error handling and state persistence.
- **Group Creation:** Users can now create groups more flexibly with options for avatars and color-coded identifiers.
- **Expense Addition:** The expense addition interface was refined to make splitting costs more straightforward, allowing itemized or equal distribution modes.

IV. Enhanced Feature: Settle Up

In addition to Iteration 1 features, the "Settle Up" function from Iteration 2 was prioritized and reintegrated with substantial improvements. It now includes:

- Smart suggestions on optimal settlement paths to minimize transaction loops.
- Visual indicators showing outstanding dues and cleared payments.
- A detailed settlement log maintained for accountability and future exports.

V. Project Development Overview

A. Team Roles

- **Kabeer Ali:** Lead Developer – responsible for rewriting the application structure and implementing the core modules.
- **Muneeb Lone:** Requirement Engineer and Developer – led the restructuring of user requirements and handled UI logic and notifications.
- **Okasha Yamin:** Developer and Tester – assisted in module development and led the testing and feedback loop.

B. Current Sprint Summary

- Reimplemented Iteration 1 features.
- Refined and readded the "Settle Up" feature.
- Set groundwork for future modules like Notifications and Export.
- Fixed adding expenses.
- Standardized UI across all pages.
- Automated debt calculation added.
- Transaction history added.

VI. User Stories

A. User Story 1: Group Creation & Management

As a user, **I want to** create a group and add members, **so that** I can manage shared expenses for specific events or social circles.

Acceptance Criteria:

- User can input a group name and optional group avatar.
- Members can be added via input fields or dropdowns.
- Group appears in the main dashboard after creation.

B. User Story 2: Add & Split Expenses

As a user, **I want to** add expenses and split them among selected group members, **so that** everyone knows how much they owe or are owed automatically.

Acceptance Criteria:

- User can select equal or custom split when entering an expense.
- Each expense records the payer and participants.
- Real-time balances update across the group dashboard.

C. User Story 3: Settle Up Debts

As a user, **I want to** log a settlement payment between members, **so that** the app updates outstanding balances and maintains a transparent payment history.

Acceptance Criteria:

- User can select a payer, payee, and amount.
- Settlement appears in transaction history.
- Balances reflect the logged payment.
- Visual indicators confirm successful debt clearance.

Application Screenshots

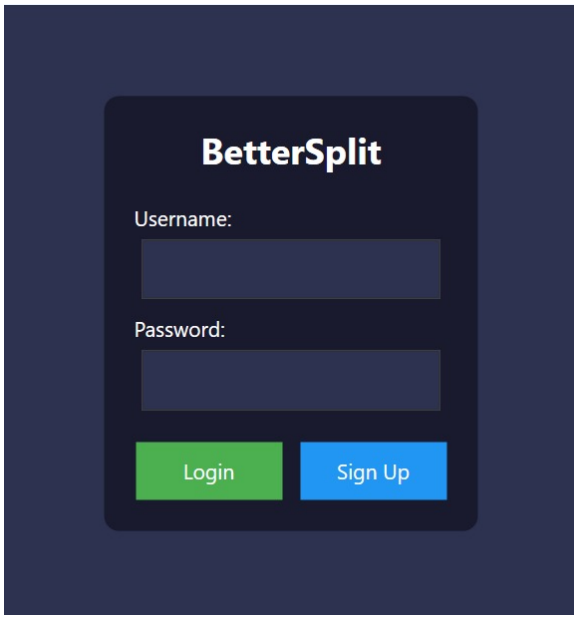


Fig. 1: Login Page

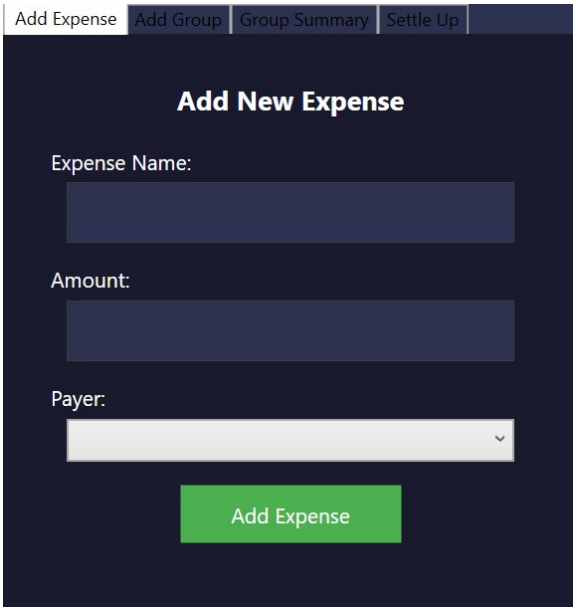


Fig. 2: Dashboard View

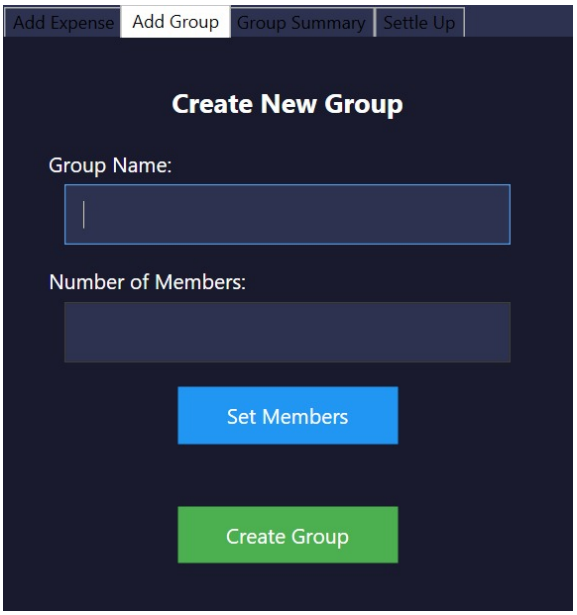


Fig. 3: Group Details

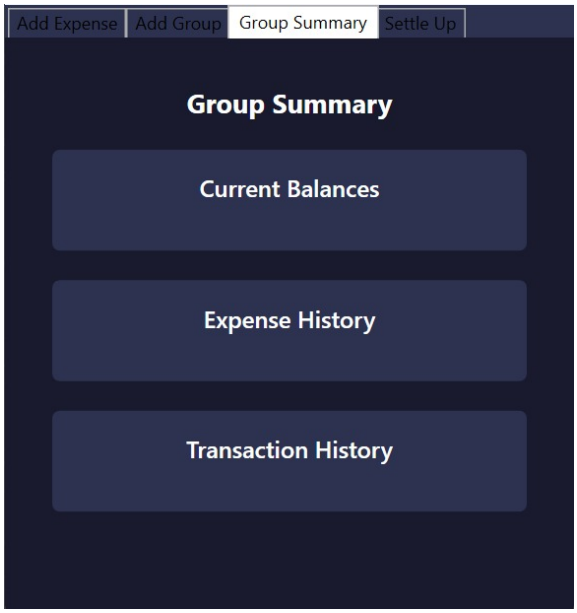


Fig. 4: Add Expense Page

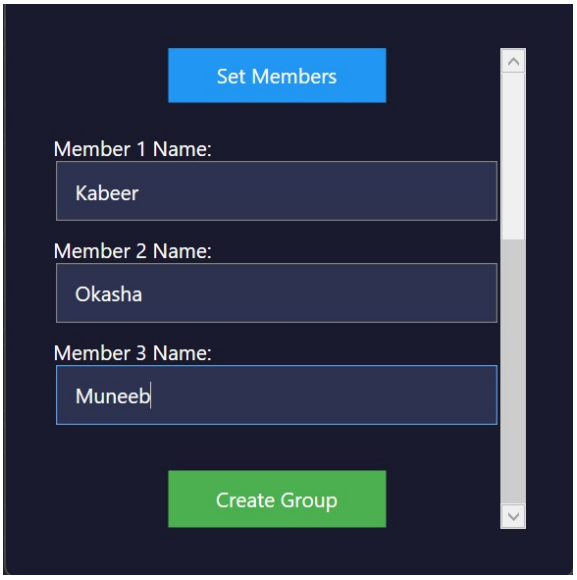


Fig. 6: Settle Up Window

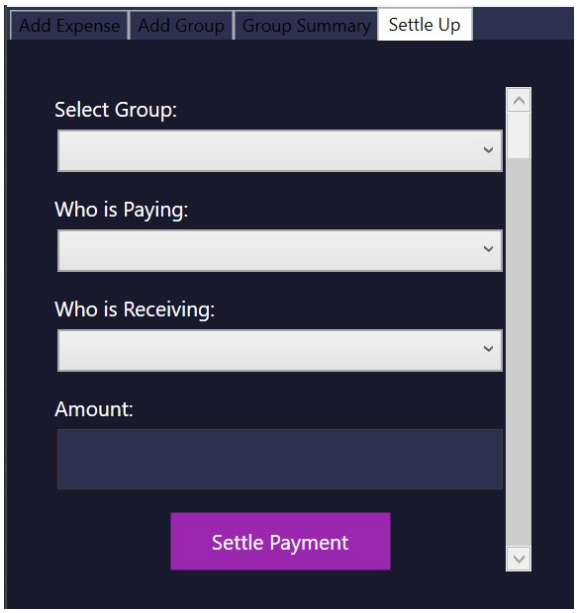


Fig. 5: Balance View

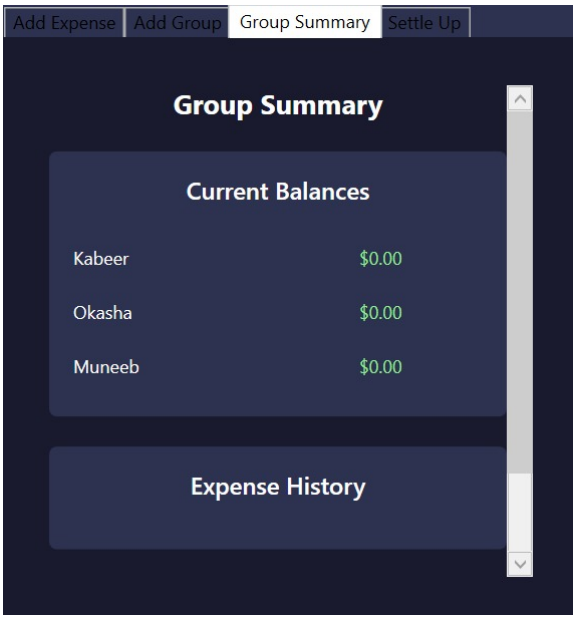


Fig. 7: Transaction History

ARCHITECTURE DIAGRAMS

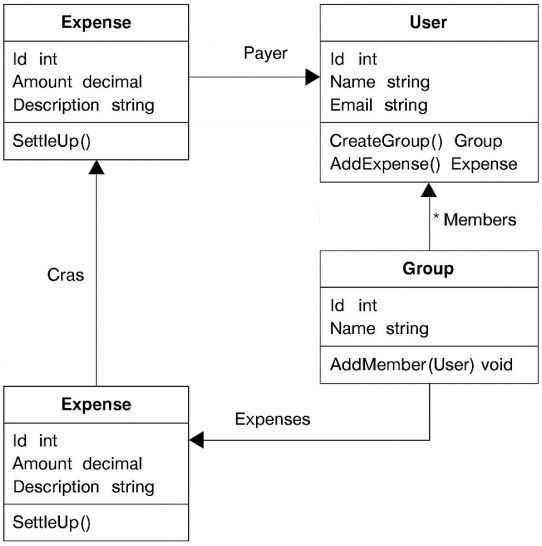


Fig. 8: MVVM Design Pattern Overview

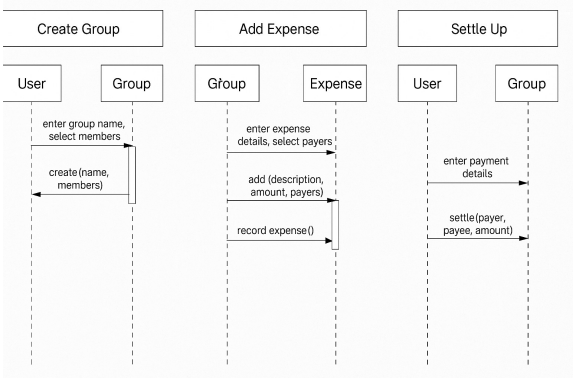


Fig. 9: Group and Expense Relationship

PROJECT MANAGEMENT (TRELLO)

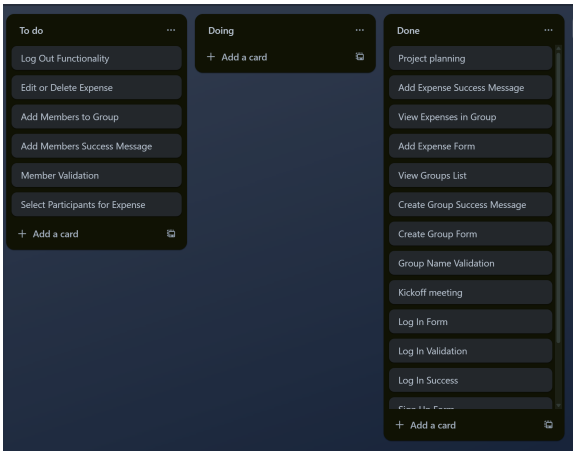


Fig. 10: Trello Board: Sprint Planning

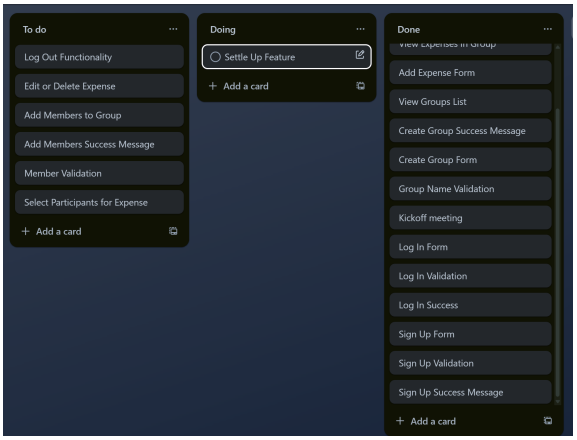


Fig. 11: Trello Board: Task Progress Overview

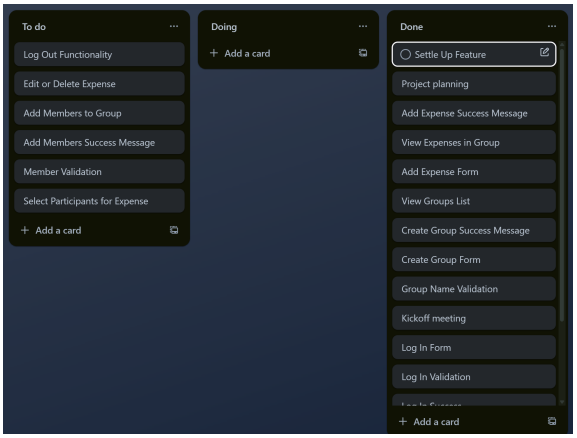


Fig. 12: Trello Board: Completed Tasks

VII. CONCLUSION AND FUTURE WORK

The rewrite has laid a strong technical foundation for BetterSplit. With core functionality now cleaner and more stable, the team is positioned to incrementally add features like Notifications, Data Export, and Real-Time Syncing. The new implementation enhances maintainability and provides a more seamless user experience.

ACKNOWLEDGMENT

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[2] A. Brown, "Effective MVVM Architecture in WPF," *Software Engineering Weekly*, vol. 15, pp. 103–110, 2021.